

Product Manual

Kamoer Fluid Tech (Shanghai) Co., Ltd.

Version : A/1

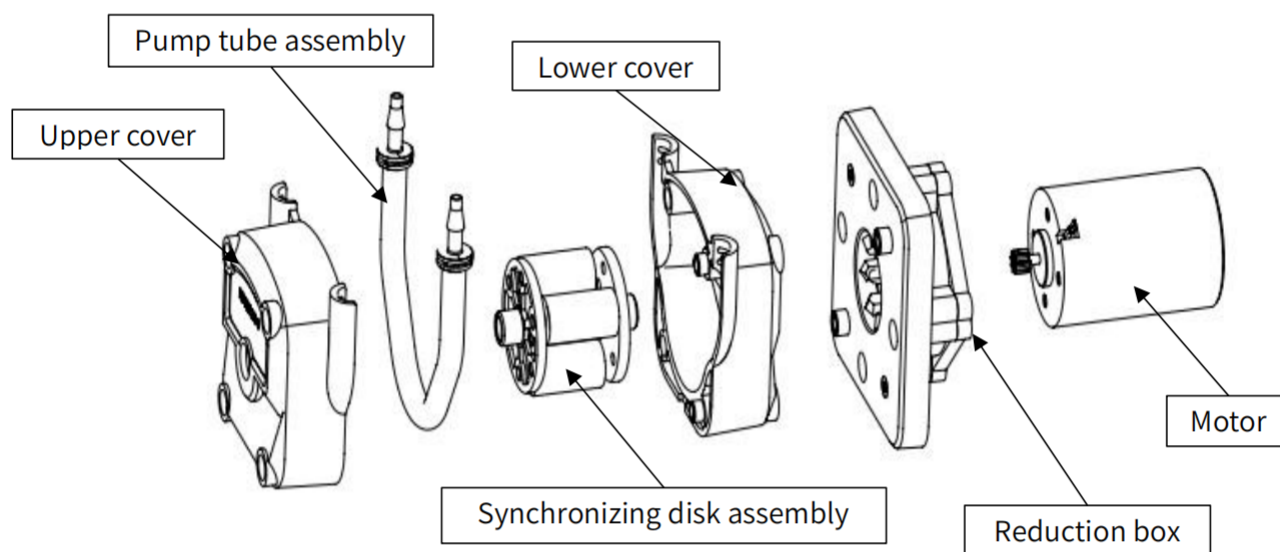
Product name	Micro peristaltic pump
Product model	KPHM600
Execution date	2022.06.13
Company name	Kamoer Fluid Tech (Shanghai) Co.,Ltd.

A. Product Overview

1. Product physical picture



2. Product assembly drawing



3. Model definition

Model	Motor model	Slow down	Cast	Tube number	Installation method
KPHM600	SV	Null	B	17	Null



3.1 Motor model



brushed motor

SW 12V DC brushed

SV 24V DC brushed



Brushless Motor

12B 12V DC brushless

24B 24V DC brushless

3.2 Deceleration

One-stage deceleration, reduction ratio 1:8

3.3 Tube type

B: BPT pump tube

3.4 Tube number

17 #: 6.4mm × 9.6mm

3.5 Installation method

Standard installation

3.6 Product material

KPHM600	PA6	POM	PTE	PTE	PP
Pump head model	Cover	Output chainring	Gear	Bracket	Connector

4. Performance characteristics, typical applications

4.1 Performance characteristics

- ◆ KPHM600 series peristaltic pump is 2 kinds of motor-driven peristaltic pump, 24V/12V motor, can provide different voltage options.
- ◆ Pump tube selection : the pump tube is imported BPT (B) tube.
- ◆ This peristaltic pump can be adapted to ordinary DC brush, DC brushless and stepper motors.
- ◆ Compact structure, large flow, high cost performance.
- ◆ Installation method: through the plate fixed, L-shaped sheet metal fixed.
- ◆ Transmission mode: gear transmission. One-stage reduction ratio 1:8. high reliability, low noise.



4.2 Typical applications

Analytical instruments: ion chromatograph, automatic car washing machine, five-category blood, hypochlorite water generator.



Ion chromatograph



Auto car wash



Five classification blood analyzer



Hypochlorite water generator

5. Product certification, intellectual property

Appearance design



Utility model



RoHS certification



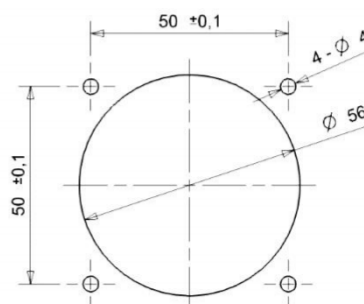
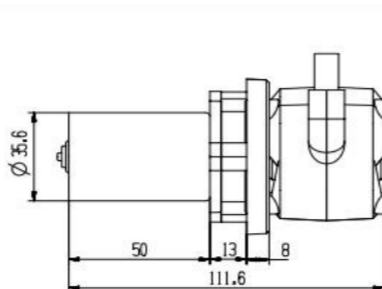
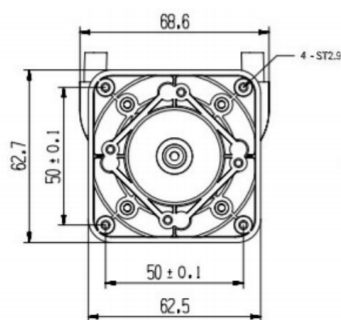
CE certification



B. Product specification

1. KPHM600 DC Brush Product Specifications

1.1 Product size



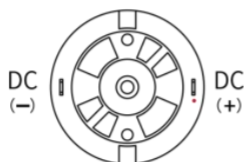
Mounting plate opening size



1.2 Technical Parameters

Project		Technical Parameters
Basic parameters	Motor type	DC brushed motor
	Flow range	$\geq 600\text{ml/min}$
	Noise	$\leq 65\text{dB}$
	Power	12W
	Control method	Switch control
	Weight	About 307g
	Rated voltage	12V/24V
	Pump tube life	BPT: 1000 hours
	Motor life	800 hours
Pump flow	Reference flow rate (unit: ml/min)	
	Pump tube code	B17 (17#)
	ID x OD	Metric: 6.4 mm $\times$ 9.6 mm Inch: 1/4 in $\times$ 3/8 in
	Pump tube material	BPT
	SV/current: 0.4A SW/Current : 0.8A	$\geq 600\text{ml/min}$
Precautions	Remark	The above parameters are measured with pure water without pressure at room temperature and standard atmospheric pressure at 20°C. Actually, there will be certain errors in the flow rate due to different media, different outlet pressures, and DC motor speed errors. The data are for reference only. In addition, it can be customized according to customer needs
	Reference noise value	In a quiet room with an ambient noise of 35dB, the product is 50cm away from the noise tester, and the noise is 62dB

1.3 Motor Control

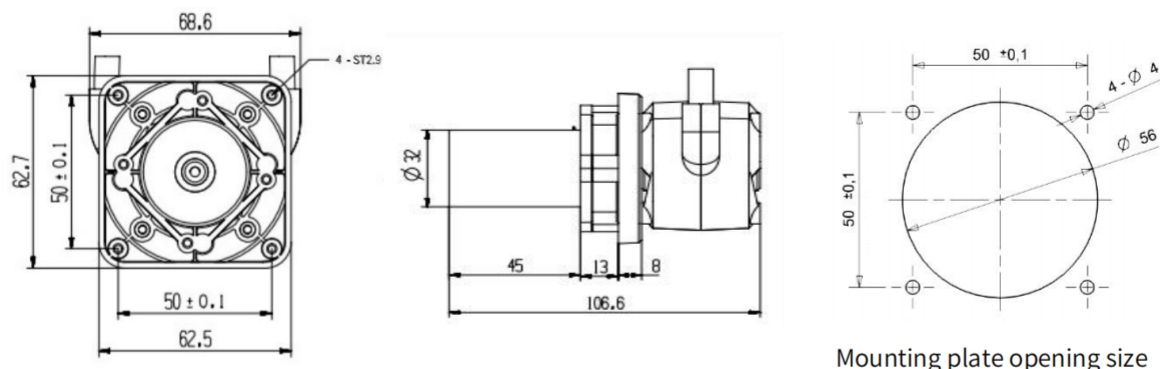


Note: According to the motor wiring diagram, the positive pole of the power supply is connected to positive, negative pole is connected to negative.



2. KPHM600 Brushless DC Product Specifications

2.1 Product Size



Mounting plate opening size

2.2 Technical parameters

Project		Technical Parameters
Basic parameters	Motor type	DC brushless motor
	Flow range	≥600ml/min
	Noise	≤65dB
	Power	12W
	Control method	Analog voltage speed regulation /PWM speed regulation
	Weight	About 242g
	Rated voltage	12V/24V
	Pump tube life	BPT: 1000 hours
	Motor life	2000 hours
Pump flow	Reference flow rate (unit: ml/min)	
	Pump tube code	B17 (17#)
	ID x OD	Metric: 6.4 mm × 9.6 mm Inch: 1/4 in × 3/8 in
	Pump tube material	BPT
	24B/current: 0.4A 12B/current: 0.8A	≥600ml/ min
Precautions	Remark	The above parameters are measured with pure water without pressure at room temperature and standard atmospheric pressure at 20°C. Actually, there will be certain errors in the flow rate due to different media, different outlet pressures, and DC motor speed errors. The data are for reference only. In addition, it can be customized according to customer needs
	Reference noise value	In a quiet room with an ambient noise of 35dB, the product is 50cm away from the noise tester, and the noise is 60dB



2.3 Description of the five-color lines of brushless motors

◆ The red wire is the positive pole of the power supply, the black wire is the negative pole of the power supply, the white wire is the speed control wire, the yellow wire is the signal feedback wire (measuring the speed), the green wire is the reversing wire (suspended to reverse, connect to the negative pole of the power supply to rotate forward).

◆ Speed control port: PWM and 0 V ~ 5V two speed control methods: ① PWM speed control, frequency 10kHz ~ 30kHz, speed control range 1 % ~ 100%, duty cycle 0 % ~ 10%, the motor does not rotate, 11 %~100% is the speed regulation range; ② 0 V ~5V voltage speed regulation, the amplitude is 5V, 0 V ~0.5V, the motor does not rotate, 0.6V~5V is the speed regulation range.

◆ Yellow line speed feedback, connected to oscilloscope probe or host computer, 1 pulse/rev, speed (rev/min)=FG signal × 60.

◆ Run at full speed, connect the red and white wires to the positive pole of the power supply, and the black wire to the negative pole.

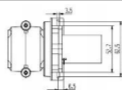
2.4 UL1332 Material: PTFE

- ◆ FG-24AWG-yellow signal feedback
- ◆ Vcc-24AWG-red positive power supply
- ◆ SP-24AWG-white speed port
- ◆ GND-24AWG-black Negative pole of power supply
- ◆ F/R-24AWG-green reversing port

C. Adapter hose

Pump tube material	Characteristic	Standards compliant	Tube number
BPT tube (B)	Resistance to common chemicals, acid and alkali and oxidation resistance life over 1000 h	RoHS FDA	17#

D. Optional accessories

Name	Specifications	Function introduction
L shape mounting plate	70.3×69.5×15mm 	Add different installation methods to meet different customer needs
Straight connector	1/4 	6.4mm×9.6mm hard tube or similar hose can be connected to the external

Special declaration: The company reserves the right to improve the appearance and technical specifications of the product without prior notice.

